Practical 1

MainActivity.java

```
package com.example.practical1helloworld;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Practical 1B: BroadcastActivity

MainActivity.java

```
package com.example.practical1brodcastactivity;

import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {
   AirPlaneModeChangeReceiver airPlaneModeChangeReceiver =
new AirPlaneModeChangeReceiver();

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);



   }

   @Override
   protected void onStart() {
      super.onStart();
      IntentFilter filter = new
IntentFilter(Intent.ACTION_AIRPLANE_MODE_CHANGED);
      registerReceiver(airPlaneModeChangeReceiver,filter);
```

```
    }

    @Override
    protected void onStop() {
        super.onStop();
        unregisterReceiver(airPlaneModeChangeReceiver);
    }
}
```

AirplaneModeChangeReceiver.java

```
package com.example.practical1brodcastactivity;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.provider.Settings;
import android.widget.Toast;

public class AirPlaneModeChangeReceiver extends
BroadcastReceiver {
    @Override
    public void onReceive(Context context, Intent intent) {
        if (isAirplaneModeOn(context.getApplicationContext())){
            Toast.makeText(context, "Airplane Mode is On",
Toast.LENGTH_SHORT).show();
        }else {
            Toast.makeText(context, "Airplane Mode is off",
Toast.LENGTH_SHORT).show();
        }
```

```
    }
    private static Boolean isAirplaneModeOn(Context context){
        return
Settings.System.getInt(context.getContentResolver(),Settings.Global.
AIRPLANE_MODE_ON,0) !=0;
    }
}
```

airplane__logo.xml

```
<vector android:height="24dp" android:tint="#000000"
    android:viewportHeight="24" android:viewportWidth="24"
    android:width="24dp" xmlns:android="http://schemas.android.com/apk/res/android">
    <path android:fillColor="@android:color/black" android:pathData="M22,16v-2l-8.5,-
5V3.5C13.5,2.67 12.83,2 12,2s-1.5,0.67 -1.5,1.5V9L2,14v2l8.5,-2.5V19L8,20.5L8,22l4,-
1l4,1l0,-1.5L13.5,19v-5.5L22,16z"/>
</vector>
```

PRACTICAL 2

Programming Resources Android Resources: (Color, Theme, String, Drawable, Dimension, Image).

Main_Activity.kt:

```
package com.rohit.drwable import
android.support.v7.app.AppCompatActivity import
android.os.Bundle class MainActivity : AppCompatActivity() {

override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)

}

}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:tools="http://schemas.android.com/tools"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".MainActivity"

    android:background="@drawable/one">
```

```
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent"/>

</LinearLayout>
```

Color.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="colorPrimary">#008577</color>
    <color name="colorPrimaryDark">#00574B</color>
    <color name="colorAccent">#D81B60</color>
</resources>
```

Style.xml

```
<resources>
    <!-- Base application theme. -->
    <style name="AppTheme" parent="Theme.AppCompat.Light.DarkActionBar">
    <!-- Customize your theme here. -->
    <item name="colorPrimary">@color/colorPrimary</item>
    <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
    <item name="colorAccent">@color/colorAccent</item>
    </style>
</resources>
```

PRACTICAL 3

Programming Activities and fragments

MainActivity.java

```
package com.example.lifecycle;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.util.Log;

public class MainActivity extends AppCompatActivity {



    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);



    }

    @Override
    protected void onStart() {
        super.onStart();
        Log.d("onStart Invoked!","on Started");
    }

    @Override
    protected void onResume() {
        super.onResume();
        Log.d("onResume Invoked!","on Resume");
    }
```

```
    @Override
    protected void onPause() {
        super.onPause();
        Log.d("onPause Invoked!","on Pause");
    }

    @Override
    protected void onStop() {
        super.onStop();
        Log.d("onStop Invoked!","on Stop");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d("onDestroy Invoked!","on Destroy");
    }
}
```

PRACTICAL 4

Programs related to different Layouts Coordinate, Linear, Relative, Table, Absolute, Frame, List View, Grid View.

1.Linear Layout

layout.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="fill_parent"
   android:layout_height="fill_parent"
   android:orientation="vertical">

   <Button
      android:id="@+id/btnStarService"
      android:layout_width="270dp"
      android:layout_height="wrap_content"
      android:layout_gravity="center"
      android:layout_margin="30dp"
      android:text="start_service" />

   <Button
      android:id="@+id/btnPauseService"
      android:layout_width="270dp"
      android:layout_height="wrap_content"
      android:layout_gravity="center"
      android:text="Pause_service"/>
   <Button
      android:id="@+id/btnStopService"
      android:layout_width="270dp"
      android:layout_height="wrap_content"
      android:layout_gravity="center"
      android:layout_margin="2px"
      android:text="Stop_service"/>
</LinearLayout>
```

MainActivity.kt

```
package com.example.myapplication

import android.os.Bundle
import android.widget.Toast
import android.widget.Button
import androidx.activity.ComponentActivity

class MainActivity : ComponentActivity() {
   override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
      setContentView(R.layout.layout)
      val btn1: Button=findViewById(R.id.StartService)
      val btn2: Button=findViewById(R.id.PauseService)
      val btn3: Button=findViewById(R.id.StopService)
      btn1.setOnClickListener { toast("Service Start")}
      btn2.setOnClickListener { toast("Service pause")}
      btn3.setOnClickListener { toast("Service Stop")}
   }
   private fun toast(message: String){
      Toast.makeText(this,message, Toast.LENGTH_SHORT).show()
   }
}
```

2.table layout

MainActivity.kt

```
package com.example.myapplication

import android.os.Bundle
import android.widget.Toast
import android.widget.Button
import androidx.activity.ComponentActivity

class MainActivity : ComponentActivity() {
   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_main)
      val btn1: Button=findViewById(R.id.btn1)
      val btn2: Button=findViewById(R.id.btn2)
      val btn3: Button=findViewById(R.id.btn3)
```

```
        val btn4: Button=findViewById(R.id.btn4)
        val btn5: Button=findViewById(R.id.btn5)
        val btn6: Button=findViewById(R.id.btn6)
        val btn7: Button=findViewById(R.id.btn7)
        val btn8: Button=findViewById(R.id.btn8)
        val btn9: Button=findViewById(R.id.btn9)
        btn1.setOnClickListener { toast("1")}
        btn2.setOnClickListener { toast("2")}
        btn3.setOnClickListener { toast("3")}
        btn4.setOnClickListener { toast("4")}
        btn5.setOnClickListener { toast("5")}
        btn6.setOnClickListener { toast("6")}
        btn7.setOnClickListener { toast("7")}
        btn8.setOnClickListener { toast("8")}
        btn9.setOnClickListener { toast("9")}
    }
private fun toast(message: String){
        Toast.makeText(this,message, Toast.LENGTH_SHORT).show()
    }
}

       /* enableEdgeToEdge()
        setContent {
            MyApplicationTheme {
                Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
                    Greeting(
                        name = "Android",
                        modifier = Modifier.padding(innerPadding)
                    )
                }
            }
        }
    }
}

@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Text(
        text = "Hello $name!",
        modifier = modifier
    )
```

```
}

@Preview(showBackground = true)
@Composable
fun GreetingPreview() {
    MyApplicationTheme {
        Greeting("Android")
    }
}*/
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TableLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="150dp"
        android:layout_marginLeft="50dp">


  <TableRow>

        <Button
            android:id="@+id/btn1"
            android:layout_gravity="center"
            android:text="1"/>
        <Button
            android:id="@+id/btn2"
            android:layout_gravity="center"
            android:text="2"/>
        <Button
            android:id="@+id/btn3"
            android:layout_gravity="center"
```

```
        android:text="3"/>
    </TableRow>


    <TableRow>
        <Button
            android:id="@+id/btn4"
            android:layout_gravity="center"
            android:text="4"/>
        <Button
            android:id="@+id/btn5"
            android:layout_gravity="center"
            android:text="5"/>
        <Button
            android:id="@+id/btn6"
            android:layout_gravity="center"
            android:text="6"/>


    </TableRow>
    <TableRow>
        <Button
            android:id="@+id/btn7"
            android:layout_gravity="center"
            android:text="7"/>
        <Button
            android:id="@+id/btn8"
            android:layout_gravity="center"
            android:text="8"/>
        <Button
            android:id="@+id/btn9"
            android:layout_gravity="center"
            android:text="9"/>
    </TableRow>
  </TableLayout>
</LinearLayout>
```

3.Relative layout

<u>layout1.xml</u>

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_height="match_parent"
    android:layout_width="match_parent"
    android:paddingTop="16dp"
    android:paddingBottom="16dp"
    android:paddingLeft="16dp"
    android:paddingRight="16dp">
    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/name"
        android:hint="@string/reminder"/>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:layout_alignParentStart="true"
        android:layout_below="@+id/name">
        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="New Button"
            android:id="@+id/button"/>
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="New Button"
        android:id="@+id/button2"/>
    </LinearLayout>
</RelativeLayout>
```

Activity_main.kt

```
package com.example.myapplication
import android.os.Bundle
import androidx.activity.ComponentActivity
class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.layout1)
    }
}
```

4.Frame Layout

layout2.xml

```
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">
   <ImageView
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:src="@drawable/sample"
      android:scaleType="centerCrop"/>
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Hello world"
      android:textSize="100dp"
      android:gravity="center"
      android:textColor="@color/teal_200"
      android:layout_marginTop="220dp"/>

</FrameLayout>
```

Activity_main.kt

```
package com.example.myapplication
import android.os.Bundle
import androidx.activity.ComponentActivity
class MainActivity : ComponentActivity() {
   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.layout2)
   }
}
```

5.List View

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
   <Button
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:id="@+id/btn"
      android:text="Click Me to see list"
      android:layout_marginLeft="110dp"
      android:layout_marginTop="250dp">

   </Button>
</LinearLayout>
```

MainActivity.kt

```
package com.example.listview

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.ListView
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat


class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
        val btn=findViewById<Button>(R.id.btn)
        btn.setOnClickListener { val intent= Intent(this,ListViewActivity::class.java) }
        startActivity(intent)
        }
    }
```

ListViewActivity,kt

```
package com.example.listview
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
class ListViewActivity : AppCompatActivity(){
   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_list_view)
   }
}
```

Strings.xml

```
<resources>
   <string name="app_name">ListView</string>
   <array name="insert_list">
      <item>one</item>
      <item>two</item>
      <item>three</item>
      <item>four</item>
      <item>five</item>
      <item>six</item>
      <item>seven</item>
      <item>eight</item>
      <item>nine</item>
      <item>ten</item>
   </array>
</resources>
```

Activity_list_view.xml

```
<ListView
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".ListViewActivity" android:entries="@array/insert_list">
```

```
</ListView>
```

6.Grid Layout

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<GridLayout
   xmlns:tools="http://schemas.android.com/tools"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity"
   android:rowCount="3"
   android:columnCount="3"
   android:padding="20dp">
   <Button
      android:layout_height="100dp"
      android:layout_width="110dp"
      android:text="1">

   </Button>
   <Button
      android:layout_height="100dp"
      android:layout_width="110dp"
      android:text="2">

   </Button>
   <Button
      android:layout_height="100dp"
      android:layout_width="110dp"
      android:text="3">

   </Button>
   <Button
      android:layout_height="100dp"
      android:layout_width="110dp"
      android:text="4">
```

```
    </Button>
    <Button
        android:layout_height="100dp"
        android:layout_width="110dp"
        android:text="5">

    </Button>
    <Button
        android:layout_height="100dp"
        android:layout_width="110dp"
        android:text="6">

    </Button>
    <Button
        android:layout_height="100dp"
        android:layout_width="110dp"
        android:text="7">

    </Button>
    <Button
        android:layout_height="100dp"
        android:layout_width="110dp"
        android:text="8">

    </Button>
    <Button
        android:layout_height="100dp"
        android:layout_width="110dp"
        android:text="9">

    </Button>


</GridLayout>
```

MainActivity.kt

```
package com.example.myapplication
```

```
import android.os.Bundle
import androidx.activity.ComponentActivity
```

PRACTICAL 5

Programming UI elements

MainActivity

```
package com.example.ui_elements;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

   Button loginbtn, registerbtn;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      Toast.makeText(this,"Main Activity
Running",Toast.LENGTH_SHORT).show();

      loginbtn = findViewById(R.id.login_btn);
      registerbtn = findViewById(R.id.register_btn);

      loginbtn.setOnClickListener(new View.OnClickListener() {
         @Override
```

```
            public void onClick(View v) {

                Toast.makeText(MainActivity.this,"Login
Clicked",Toast.LENGTH_SHORT).show();

                Intent intent = new Intent(MainActivity.this, LoginActivity.class);
                startActivity(intent);
            }
        });

        registerbtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                Toast.makeText(MainActivity.this,"Register
Clicked",Toast.LENGTH_SHORT).show();

                Intent intent = new Intent(MainActivity.this, RegisterActivity.class);
                startActivity(intent);
            }
        });
    }
}
```

LoginActivity

```
package com.example.ui_elements;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.widget.Toast;

public class LoginActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);

        Toast.makeText(this,"Login Activity Started",Toast.LENGTH_SHORT).show();

        setContentView(R.layout.activity_login);
    }

    }
```

RegisterActivity

```
package com.example.ui_elements;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

public class RegisterActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_register);
    }
}
```

activity_Register.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
```

```
    tools:context=".RegisterActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="REGISTER"
        android:textSize="30dp"
        android:fontFamily="@font/akaya_telivigala"
        android:layout_centerInParent="true"
        android:textColor="@color/white"/>

</RelativeLayout>
```

activity_Login.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="LOGIN PAGE"
        android:textSize="25sp" />

</LinearLayout>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="@drawable/img"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/channelname"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:fontFamily="@font/akaya_telivigala"
        android:text="Code with yash"
        android:textColor="@color/white"
        android:layout_marginTop="200dp"
        android:textSize="50dp" />

    <Button
        android:id="@+id/login_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@+id/channelname"
        android:layout_margin="20dp"
        android:text="Login"
        android:textAllCaps="true"
        android:textSize="25dp"
        android:textAlignment="center"
        android:fontFamily="@font/akaya_telivigala"/>

<Button
        android:id="@+id/register_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
        android:layout_below="@+id/login_btn"
        android:layout_margin="20dp"
        android:text="Register"
        android:textAllCaps="true"
        android:textSize="25dp"
        android:textAlignment="center"
        android:fontFamily="@font/akaya_telivigala"/>



</RelativeLayout>
```

PRACTICAL 6

Programming menus, dialog, dialog fragments

MainActivity.java

```
package com.example.practical6;

import android.app.AlertDialog;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.SeekBar;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnDialog;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnDialog = findViewById(R.id.btnDialog);

        btnDialog.setOnClickListener(v -> showBrightnessDialog());
    }

    private void showBrightnessDialog() {
```

```
        LayoutInflater inflater = getLayoutInflater();
        View view = inflater.inflate(R.layout.brightness, null);

        CheckBox checkAuto = view.findViewById(R.id.checkAuto);
        SeekBar seekBar = view.findViewById(R.id.seekBrightness);

        AlertDialog.Builder builder = new AlertDialog.Builder(this);
        builder.setView(view);

        builder.setPositiveButton("OK", (dialog, which) -> {
            int value = seekBar.getProgress();
            boolean auto = checkAuto.isChecked();

            Toast.makeText(this,
                    "Brightness: " + value + " Auto: " + auto,
                    Toast.LENGTH_SHORT).show();
        });

        builder.setNegativeButton("Cancel", (dialog, which) -> dialog.dismiss());

        builder.show();
    }
}
```

practical 6B

manu_option

MainActivity.java

```
package com.example.menu_options;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // return super.onCreateOptionsMenu(menu);
        getMenuInflater().inflate(R.menu.menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        int id = item.getItemId();
        if (id == R.id.item1) {
```

```
            Toast.makeText(this, "Item 1 Selected", Toast.LENGTH_SHORT).show();
        } else if (id == R.id.item2) {
            Toast.makeText(this, "Item 2 Selected", Toast.LENGTH_SHORT).show();
        } else if (id == R.id.item3) {
            Toast.makeText(this, "Item 3 Selected", Toast.LENGTH_SHORT).show();
        } else {
            return super.onOptionsItemSelected(item);
        }

        return true;
    }
}
```

menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:android="http://schemas.android.com/apk/res/android">

    <item android:title="Item1"
        android:id="@+id/item1"/>
    <item android:title="Item2"
        android:id="@+id/item2"/>
    <item android:title="Item3"
        android:id="@+id/item3"/>
</menu>
```

Values/themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Menu_Options"
parent="Theme.AppCompat.DayNight.DarkActionBar">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.Menu_Options" parent="Base.Theme.Menu_Options" />
</resources>
```

Night/themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
   <!-- Base application theme. -->
   <style name="Base.Theme.Menu_Options"
parent="Theme.AppCompat.DayNight.DarkActionBar">
      <!-- Customize your dark theme here. -->
      <!-- <item name="colorPrimary">@color/my_dark_primary</item> -->
   </style>
</resources>
```

Practical 8

Programs on Services, notification and broadcast receivers

MainActivity.java

```
package com.example.practical8;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

MyReceiver.java

```
package com.example.practical8;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.widget.Toast;

public class MyReceiver extends BroadcastReceiver {

    @Override
    public void onReceive(Context context, Intent intent) {

        if (Intent.ACTION_AIRPLANE_MODE_CHANGED.equals(intent.getAction())) {

            Toast.makeText(context,ndroid
```

```
            "Broadcast: Flight mode changed.",
            Toast.LENGTH_LONG).show();
        }
    }
}
```

AndroidManifest.xml (not necessary written only understanding the code)

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@drawable/flight_mode"
        android:label="@string/app_name"
        android:roundIcon="@drawable/flight_mode"
        android:supportsRtl="true"
        android:theme="@style/Theme.Practical8"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <receiver
            android:name=".MyReceiver"
            android:enabled="true"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.AIRPLANE_MODE"/>
            </intent-filter>

        </receiver>
    </application>
```

```
</manifest>
```

Flight_mode.xml

```
<vector android:height="24dp" android:tint="#000000"
    android:viewportHeight="24" android:viewportWidth="24"
    android:width="24dp"
xmlns:android="http://schemas.android.com/apk/res/android">
    <path android:fillColor="@android:color/black"
android:pathData="M22,16v-2l-8.5,-5V3.5C13.5,2.67 12.83,2 12,2s-
1.5,0.67 -1.5,1.5V9L2,14v2l8.5,-2.5V19L8,20.5L8,22l4,-1l4,1l0,-
1.5L13.5,19v-5.5L22,16z"/>
</vector>
```

Colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
</resources>
```

PRACTICAL 9

Database Programming with SQLite

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:padding="20dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="SQLite Tutorial - User Management"
      android:textSize="18sp"
      android:textStyle="bold"/>

   <EditText
      android:id="@+id/etUserId"
      android:hint="User ID"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"/>

   <EditText
      android:id="@+id/etName"
      android:hint="User Name"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"/>

   <EditText
      android:id="@+id/etAge"
```

```
    android:hint="User Age"
    android:inputType="number"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<LinearLayout
    android:layout_marginTop="10dp"
    android:orientation="horizontal"
    android:layout_width="match_parent"
    android:layout_height="wrap_content">

    <Button
        android:id="@+id/btnAdd"
        android:text="ADD"
        android:layout_weight="1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnDelete"
        android:text="DELETE"
        android:layout_weight="1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnShow"
        android:text="SHOW ALL"
        android:layout_weight="1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"/>
</LinearLayout>

<TextView
    android:id="@+id/tvResult"
```

```
        android:layout_marginTop="20dp"
        android:textSize="16sp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.sqlitetutorial;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etUserId, etName, etAge;
    Button btnAdd, btnDelete, btnShow;
    TextView tvResult;
    DatabaseHelper db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etUserId = findViewById(R.id.etUserId);
        etName = findViewById(R.id.etName);
        etAge = findViewById(R.id.etAge);
        btnAdd = findViewById(R.id.btnAdd);
        btnDelete = findViewById(R.id.btnDelete);
```

```
        btnShow = findViewById(R.id.btnShow);
        tvResult = findViewById(R.id.tvResult);

        db = new DatabaseHelper(this);

        btnAdd.setOnClickListener(v -> {
            boolean inserted = db.addUser(
                    etUserId.getText().toString(),
                    etName.getText().toString(),
                    etAge.getText().toString());

            if (inserted)
                Toast.makeText(this, "User Added", Toast.LENGTH_SHORT).show();
            else
                Toast.makeText(this, "Error Adding User",
Toast.LENGTH_SHORT).show();
        });

        btnDelete.setOnClickListener(v -> {
            boolean deleted = db.deleteUser(
                    etUserId.getText().toString());

            if (deleted)
                Toast.makeText(this, "User Deleted", Toast.LENGTH_SHORT).show();
            else
                Toast.makeText(this, "User Not Found", Toast.LENGTH_SHORT).show();
        });

        btnShow.setOnClickListener(v ->
                tvResult.setText(db.getAllUsers()));
    }
}
```

DatabaseHelper.java

```
package com.example.sqlitetutorial;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "UserDB";
    private static final int DATABASE_VERSION = 1;

    private static final String TABLE_NAME = "users";
    private static final String COL_ID = "userid";
    private static final String COL_NAME = "name";
    private static final String COL_AGE = "age";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String CREATE_TABLE = "CREATE TABLE " + TABLE_NAME +
                " (" + COL_ID + " TEXT PRIMARY KEY, "
                + COL_NAME + " TEXT, "
                + COL_AGE + " TEXT)";
        db.execSQL(CREATE_TABLE);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

    public boolean addUser(String id, String name, String age) {
        SQLiteDatabase db = this.getWritableDatabase();
```

```
        ContentValues values = new ContentValues();
        values.put(COL_ID, id);
        values.put(COL_NAME, name);
        values.put(COL_AGE, age);

        long result = db.insert(TABLE_NAME, null, values);
        return result != -1;
    }

    public boolean deleteUser(String id) {
        SQLiteDatabase db = this.getWritableDatabase();
        int result = db.delete(TABLE_NAME, COL_ID + "=?", new String[]{id});
        return result > 0;
    }

    public String getAllUsers() {
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.rawQuery("SELECT * FROM " + TABLE_NAME, null);

        StringBuilder buffer = new StringBuilder();

        while (cursor.moveToNext()) {
            buffer.append(cursor.getString(0)).append(" - ")
                    .append(cursor.getString(1)).append(" - ")
                    .append(cursor.getString(2)).append("\n");
        }

        cursor.close();
        return buffer.toString();
    }
}
```